6 Postcolonial Perspectives

GEB2508 AI and Society

Review of Last Class

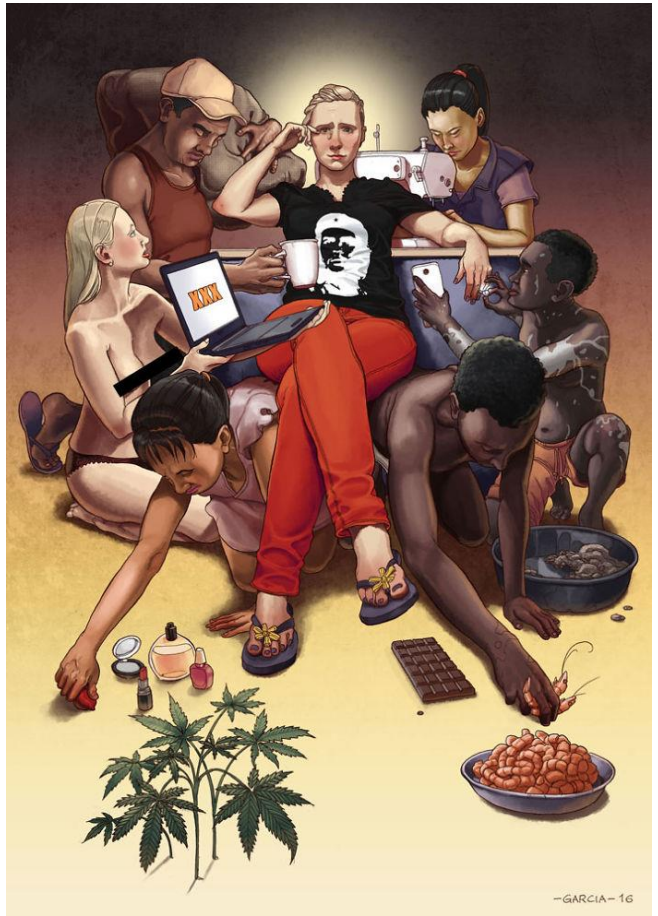
- Feminism: Theory & Reality
  - Sex and Gender
  - Determinism vs. Constructivism
  - Gender Inequality
- Feminist Perspectives on Science and Technology
  - Gender Representation and Normativity in STEM
  - Gendered Scientific Knowledge
  - Gendered Technology

Content

- Empire, Colonialism, and Orientalism
  - Scientific Racism
- Science and Technology as Colonial Infrastructures
  - S&T as Tools of Empire
  - Empire as A Laboratory for S&T
- Postcolonial Perspectives of AI
  - Data Colonialism
  - Cultural Hegemony
  - Global Labor Regime & Supply Chain

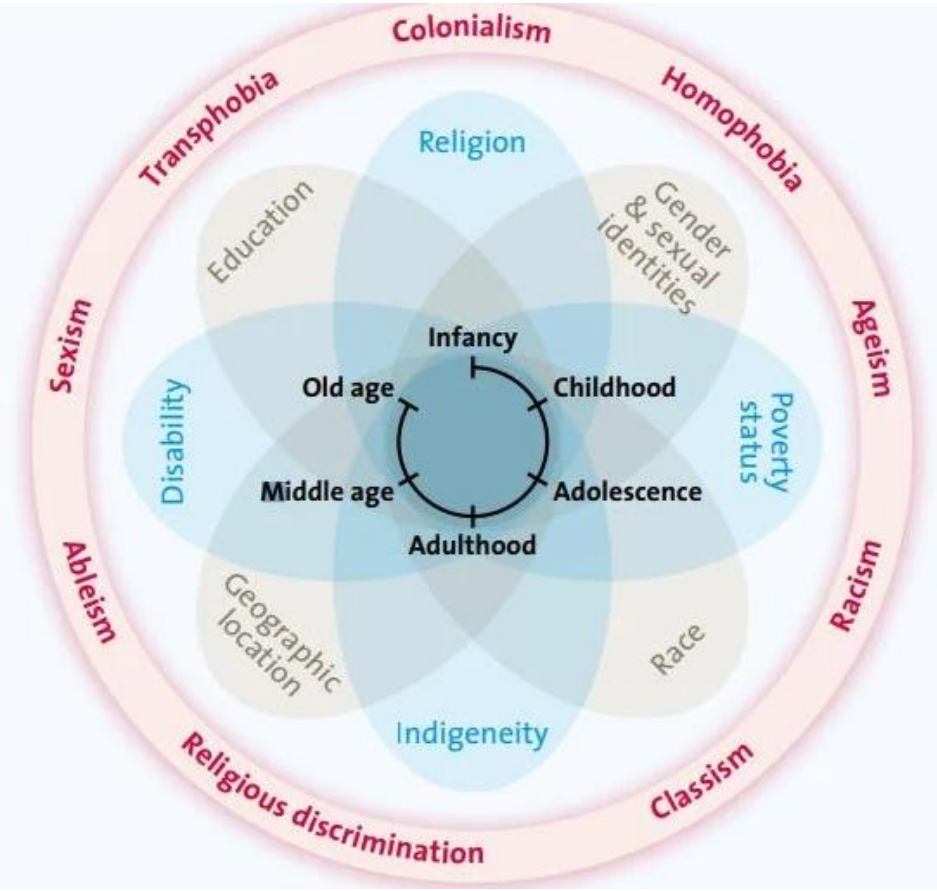
Critiques on Marxism

- Gender and race blindness - recognizes **only class**-based social groupings
- Economic reductionism and determinism
- Feminist critique: separation of production and reproduction; undervaluation of **non-market** labor
- Postcolonial critique: Eurocentrism



Intersectionality

- Intersectionality: the complex and cumulative way in which **multiple** forms of discrimination combine, overlap, or intersect
- Interlocking systems of discrimination and privilege: class is not the only axis of oppression; systems of domination are interconnected and cannot be fully understood in isolation.

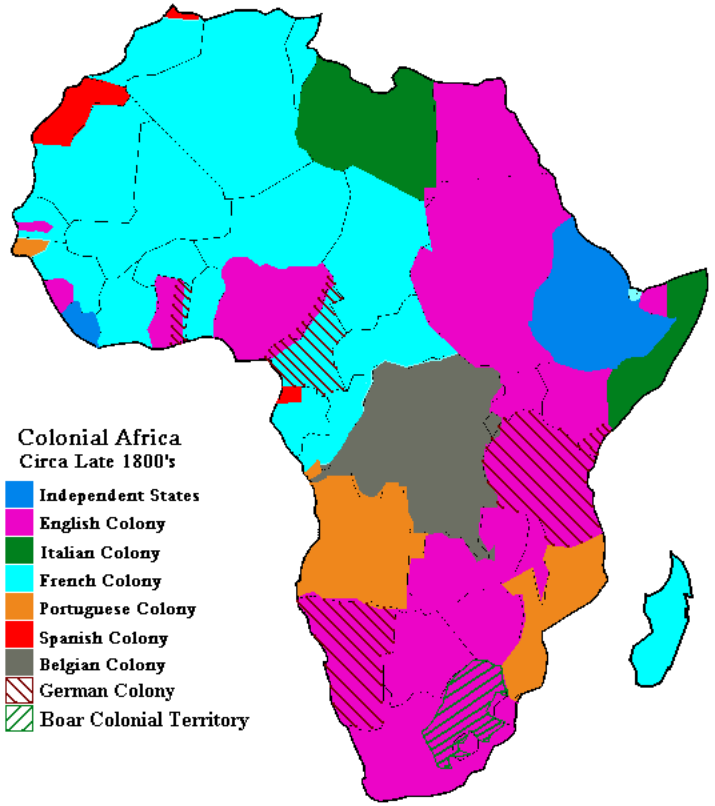


Colonialism

- The Industrial Revolution created a demand for resources and new markets to sell their goods.
- Colonialism involved the physical occupation and full or partial **political** control over another country, accompanied by **economic** exploitation.
  - Culture, institutions, and rules
  - Economic dependence

Scramble for Africa

- Transatlantic slave trade as the largest **forced** migration in human history
- Rapid colonial expansion: European control in Africa grew from ~10% (1870) to ~90% (1914).
- Technology as the key driver: medicine, military, railroads, telegraph, etc.
- Berlin Conference (1884-1885)
  - Europeans divided the African continent without considering the native groups.



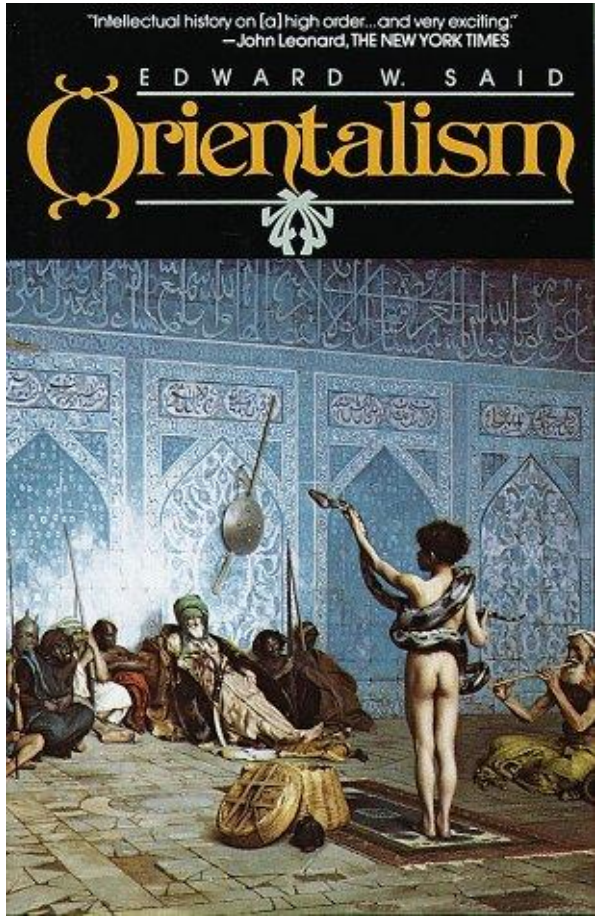
Postcolonialism

- The critical academic study of the enduring cultural and socio-political **legacy** of colonialism.
- Foreground colonized perspectives and recover the **untold** stories of the oppressed
- Deconstruct existing structures of power and knowledge

Thanks-giving vs. Thanks-taking

Orientalism

- Etymology: terms like “Orient,” “Middle East,” and “Far East” reflect **Eurocentric** assumptions.
- Orientalism is a postcolonial framework for analyzing how knowledge production **sustains** unequal power relations
  - Knowledge that constructed the “Orient” as exotic, backward, and morally inferior
  - These representations **justified** colonial domination



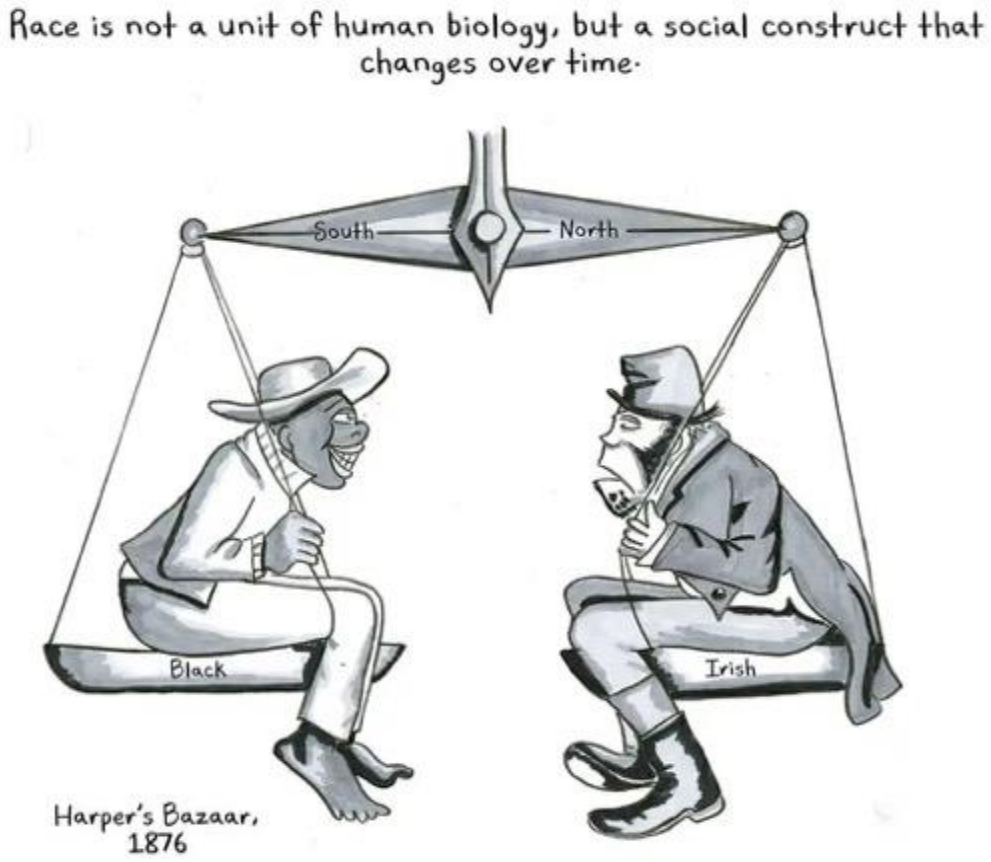
Colonial Gaze

Edouard Manet, Olympia (1863)

Gone with the Wind (1939)

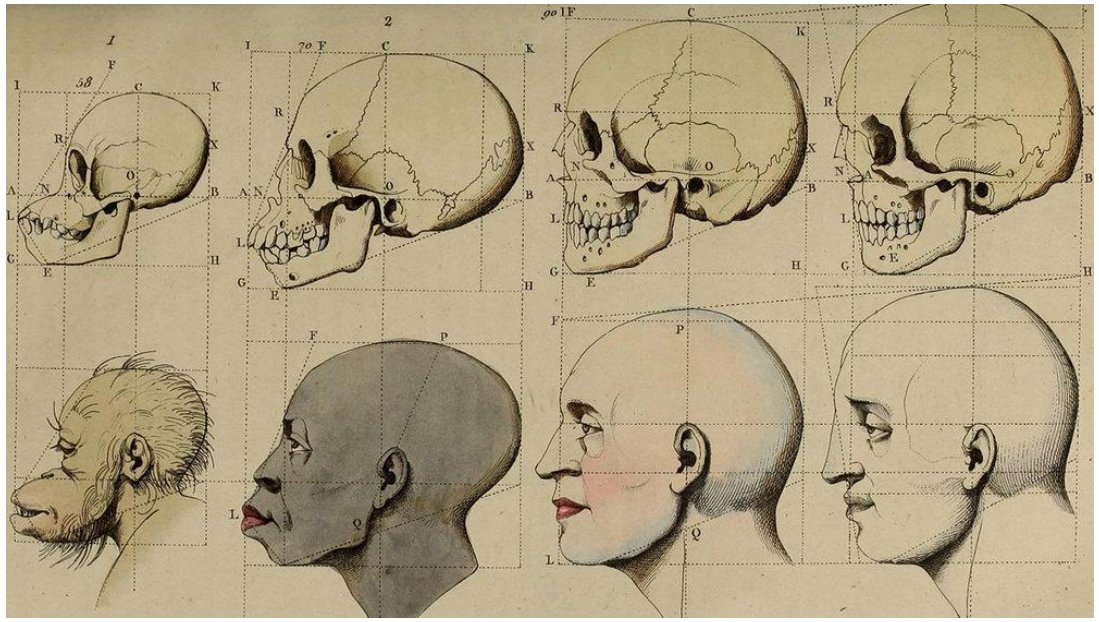
Social Construction of Race

- Race is a social construct designed to **justify** hierarchies and inequalities.
  - Colonial expansion and the trans-Atlantic slave trade
- Rise of scientific racism
- Genetic research rejects the existence of any **biological** or **scientific basis** of race



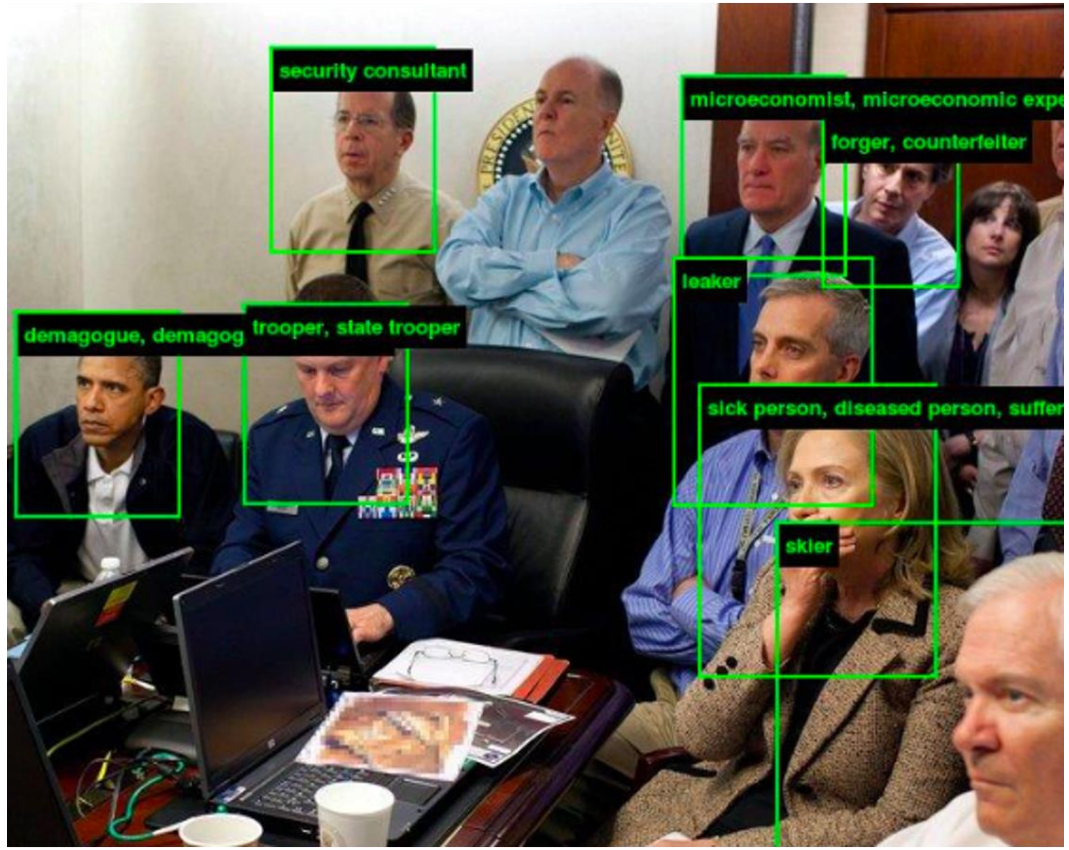
Scientific Racism

- The pseudoscience employed to construct a racial hierarchy and **justify** racism
  - Anthropometry (measurement of human sizes and forms)
  - Craniometry (measurement of human skulls)
  - Phrenology (study of the contours of human skulls)



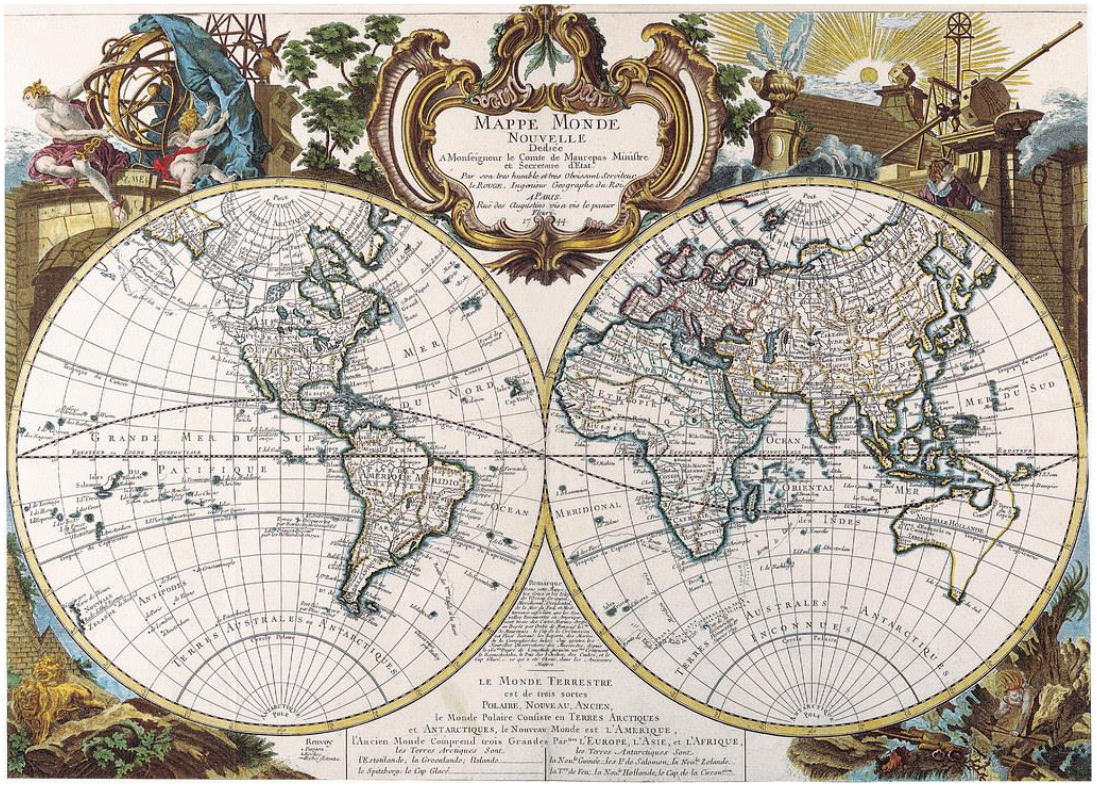
Implications for AI: Politics of Classification

- Lessons from scientific racism: biased ideological assumptions produce ‘scientific’ findings which then legitimize those biases, creating a **self-reinforcing cycle** of knowledge and power.
- Classification as **social ordering**
  - Amazon’s AI recruiting tool
  - Derogatory labels of ImageNet



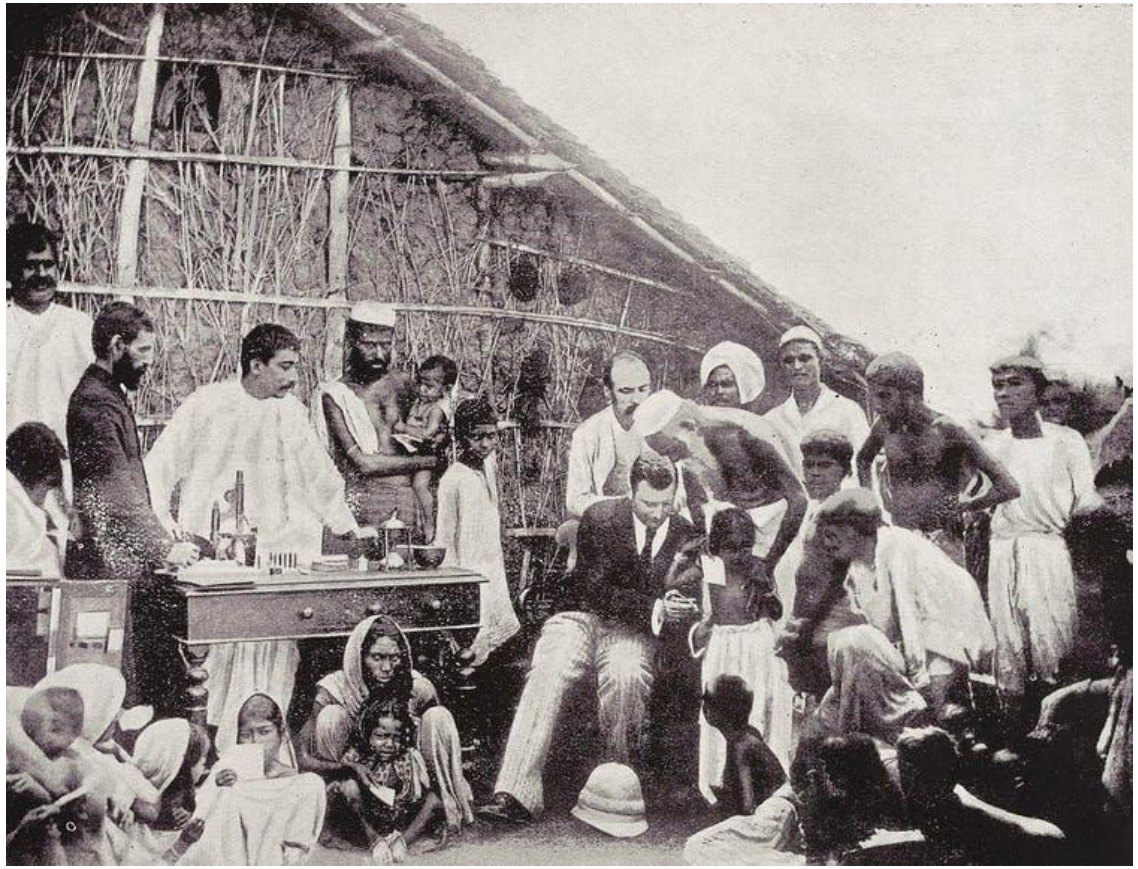
S&T as A Tool of Empire

- Exploration and navigation
  - Navigation and shipbuilding
  - Geography and cartography
- Natural history and resource exploitation
  - Botany and geology
- Medicine and public health
  - Colonial medicine
  - Epidemiology



The Case of Medicine

- Colonial medicine and **productivity**
- Medicine as moral **justification**: evidence of the empire's benevolence
- Tropical diseases attributed to the "uncivilized and unclean" local practices
- Colonization of the body



Empire as A Laboratory for S&T

- Colonized regions provided specimens for the empire, advancing fields like botany, ecology, and geology.
- Colonies served as **living laboratories** for agricultural and testing grounds for medical experiments
- Universities as colonial knowledge institutions

Colonial Science

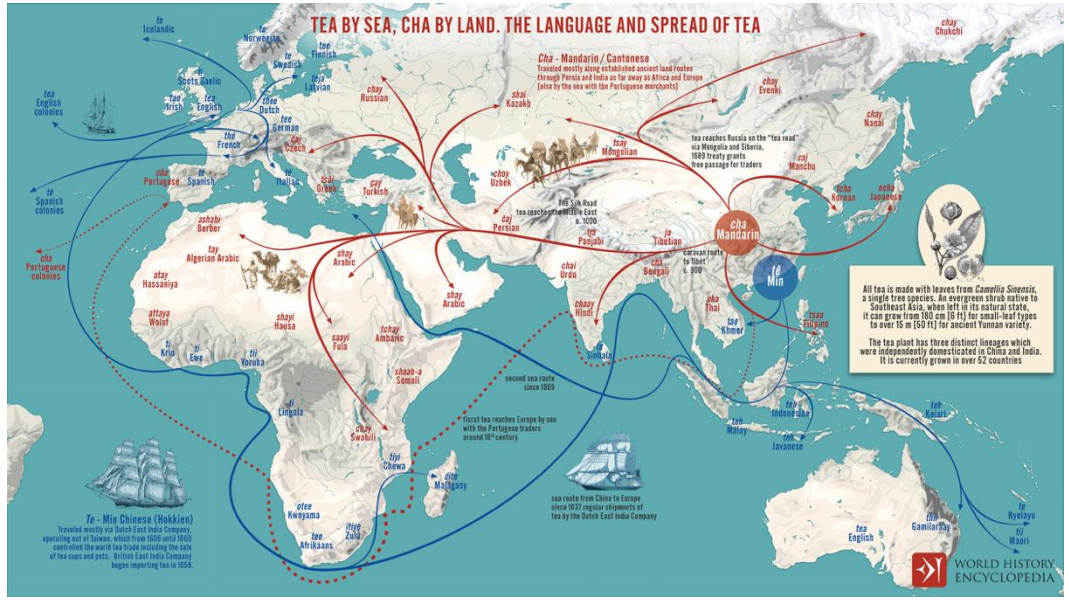
- Institutions like the British Museum and the Natural History Museum depended on a global network of colonial collectors and transnational supply chains
- Royal Botanic Gardens at Kew functioned as a hub for global botanical research and imperial resource management
- “Collected” or “stolen”

The Case of Botany

- Botanists as empire's agents
- Cash-crop regimes as a form of colonial **violence**
  - Plantation agriculture reshaped landscapes and disrupted local ecologies.
- Re-classifying and **reordering** nature
  - Erasure of indigenous knowledge and practices

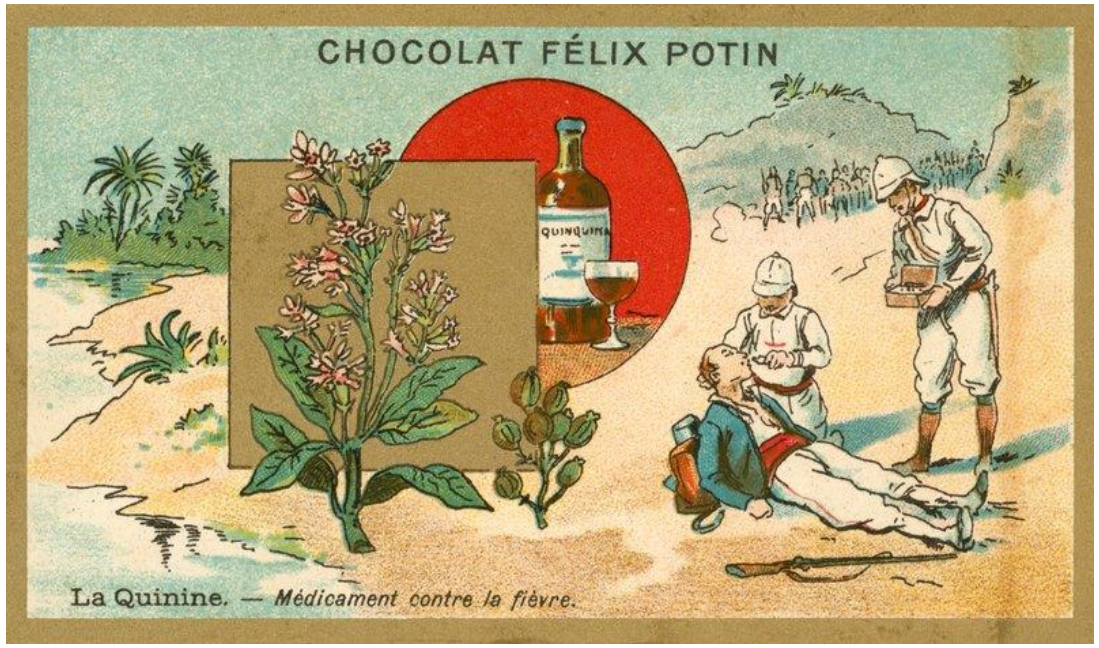
The Case of Tea

- The British East India Company sent Robert Fortune to steal the secrets of tea horticulture and manufacturing in 1848.
- "There was a time when maps of the world were redrawn in the name of plants, when two empires, Britain and China, went to war over two flowers: the poppy and the camellia." (*For All the Tea in China*)



Colonial Knowledge Appropriation

- Center-periphery flow: indigenous knowledge was extracted and reframed as Western science.
- Indigenous peoples were regarded as **sources** of information rather than **contributors** to the advancement of science.
  - Quinine & smallpox inoculation
- Double movement: extraction of useful knowledge & devaluation of its knowledge systems

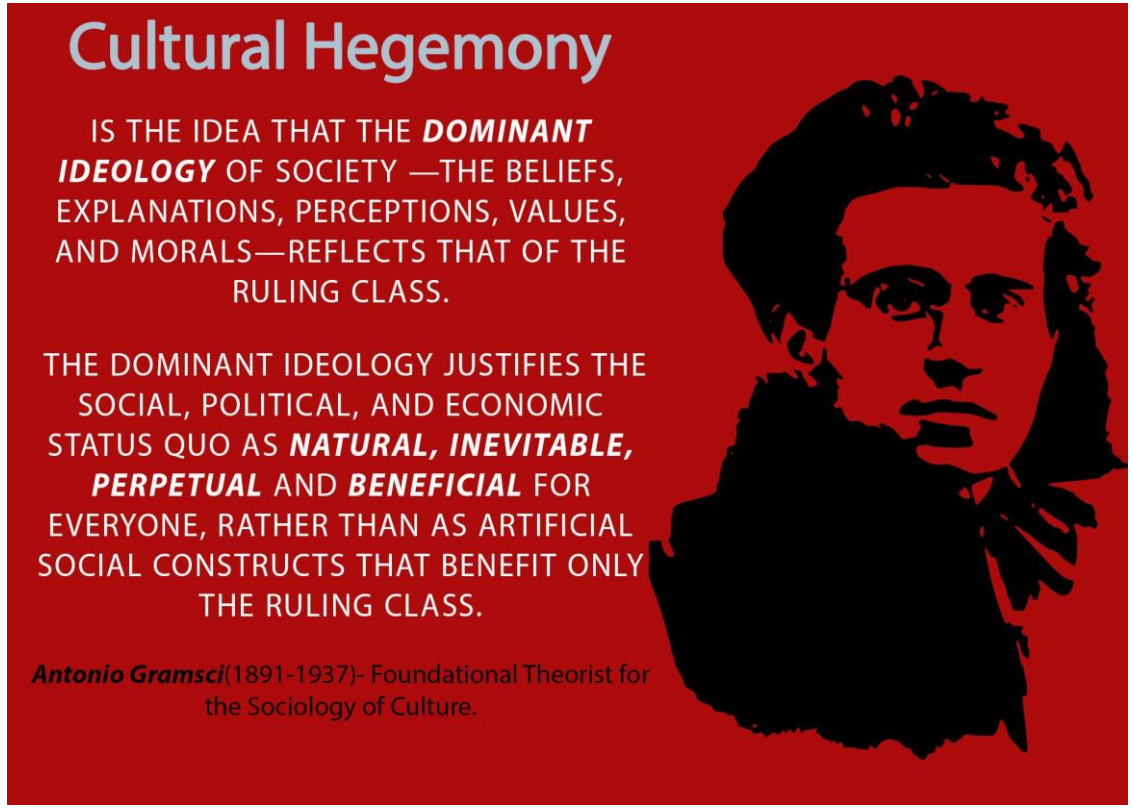


Data Colonialism

- Data as the new oil
  - Resource extraction → data extraction
- Global **asymmetries** of data flows
  - Colonization → datafication
  - **Value** flows outward while **dependency** flows inward
- Platforms as imperial **infrastructures**
  - Old routes (ports, canals, and railways) → New networks (social media networks, telecom providers, and cloud computing services)

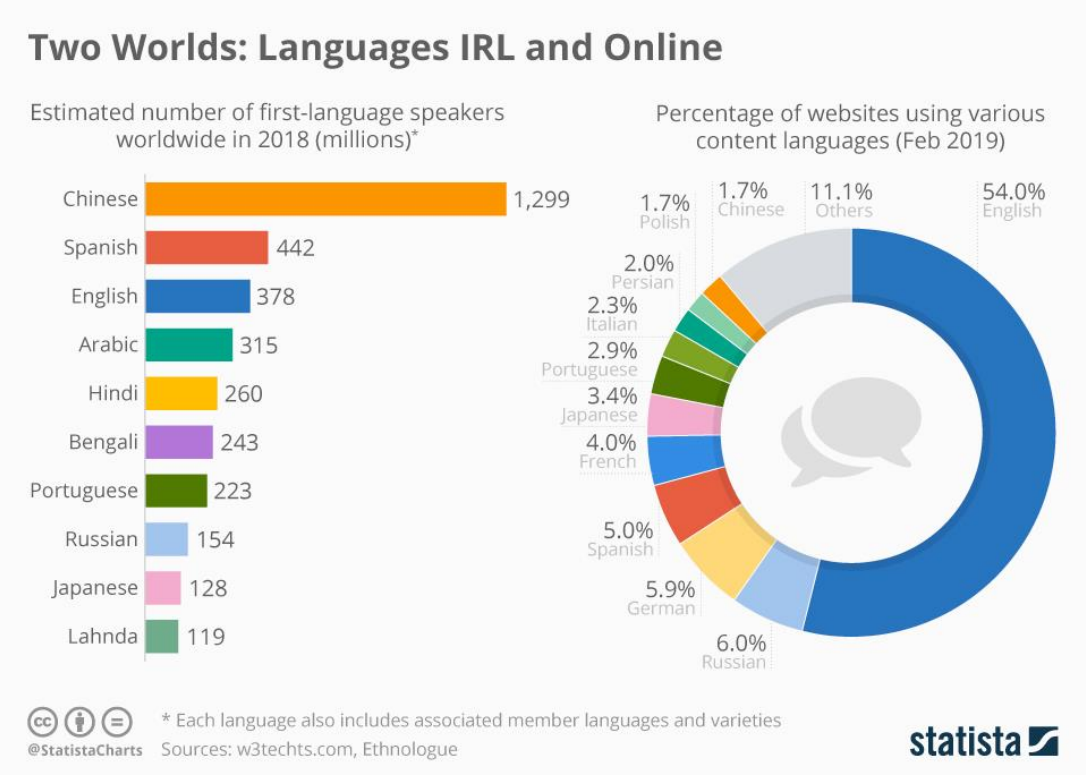
Cultural Hegemony

- Antonio Gramsci (1891-1937)
- Coercion vs. consent
- Cultural hegemony: ruling-class values presented as natural, good, and universal
- Politics as **cultural** struggle: prolonged “war of position” through contests over meanings and values



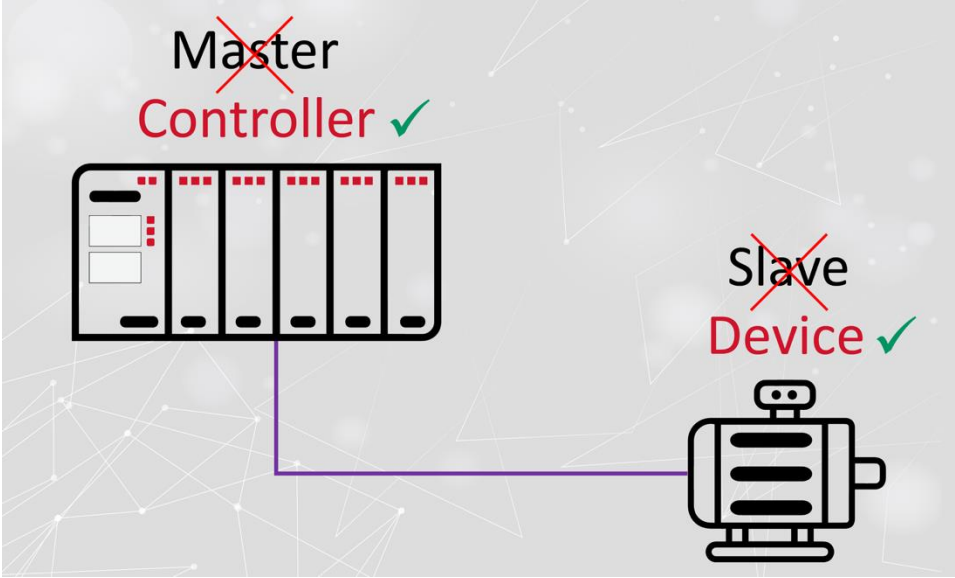
Cultural Hegemony Through AI

- Western **linguistic** dominance in AI systems
  - NLP bias toward English
- Cultural representation
  - Stereotyping and exoticism
- Platform **cultural imperialism**
  - Whose norms for content moderation & recommendation
  - What counts as acceptable humor, blasphemy, or satire



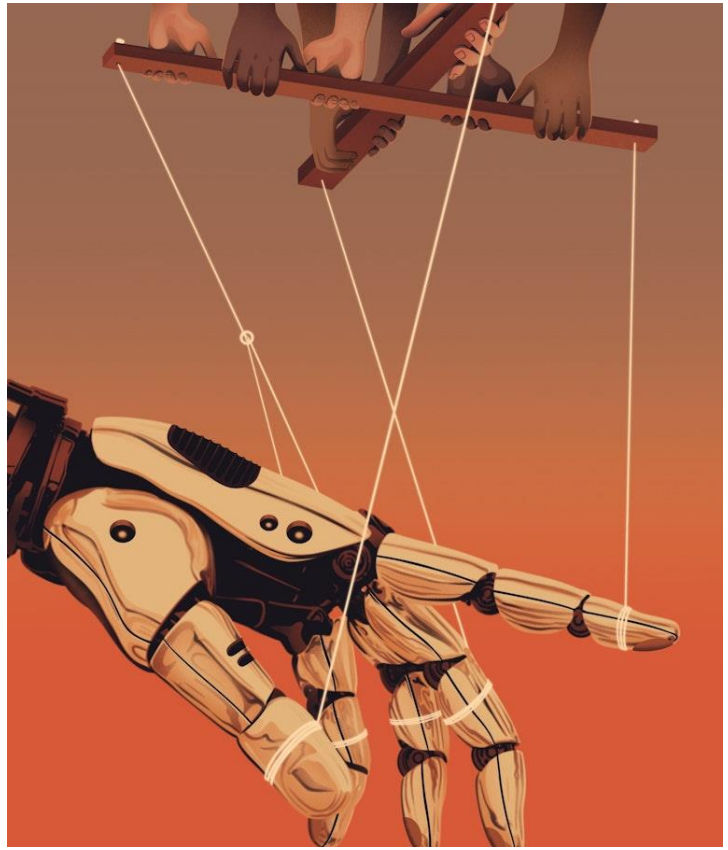
History of “Master-Slave” Terminology

- Use of the terminology in engineering and computing
  - Hardware, robotics, circuits
- Non-neutral metaphors do cognitive work
  - Obscure **historical** violence and normalize **unequal** power relations
- Criticism and reconsideration
  - Alternatives: primary/replica, leader/follower, main/secondary



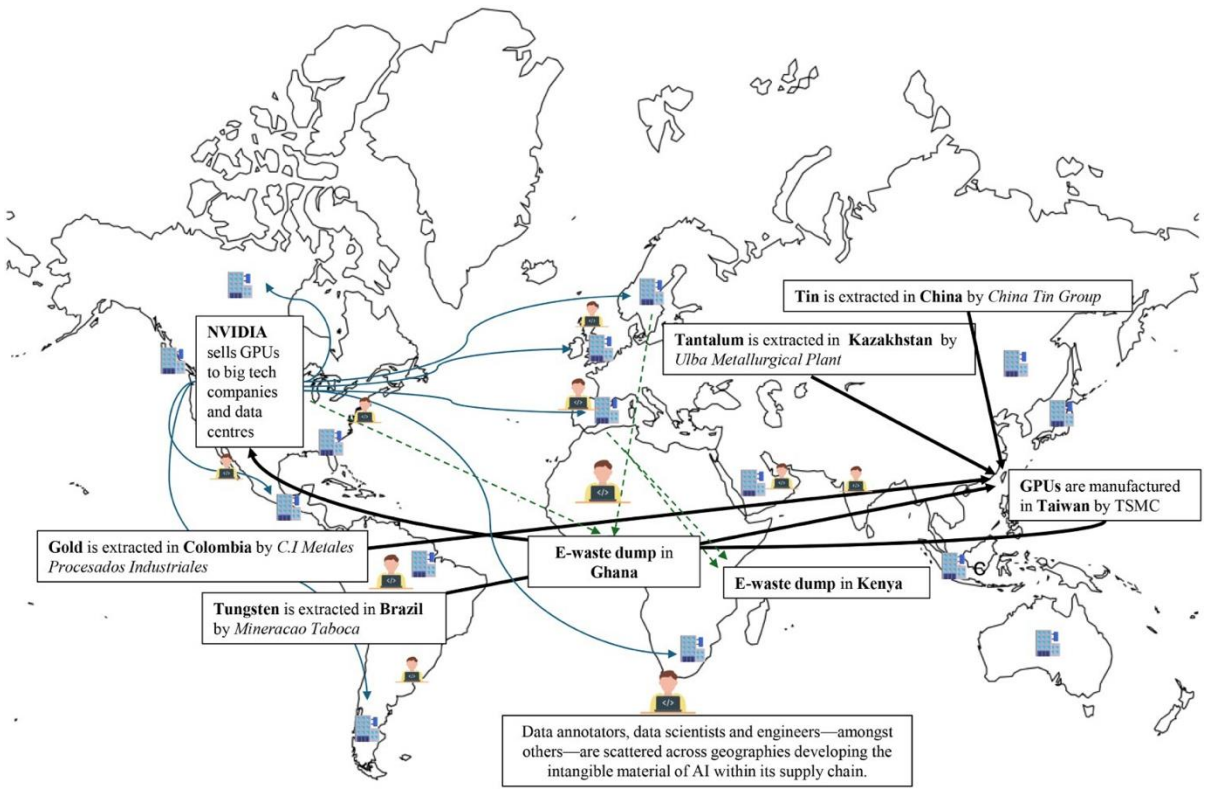
AI's Global Labor Regime

- Global division of labor in AI reproduces old colonial hierarchies
  - High-wage design and R&D in global North
  - Low-wage labeling and content moderation in global South
- Platformization as new forms of dependency
  - Content moderation in Kenya & Philippines
  - Data extraction in African telecom markets



AI's Global Supply Chain

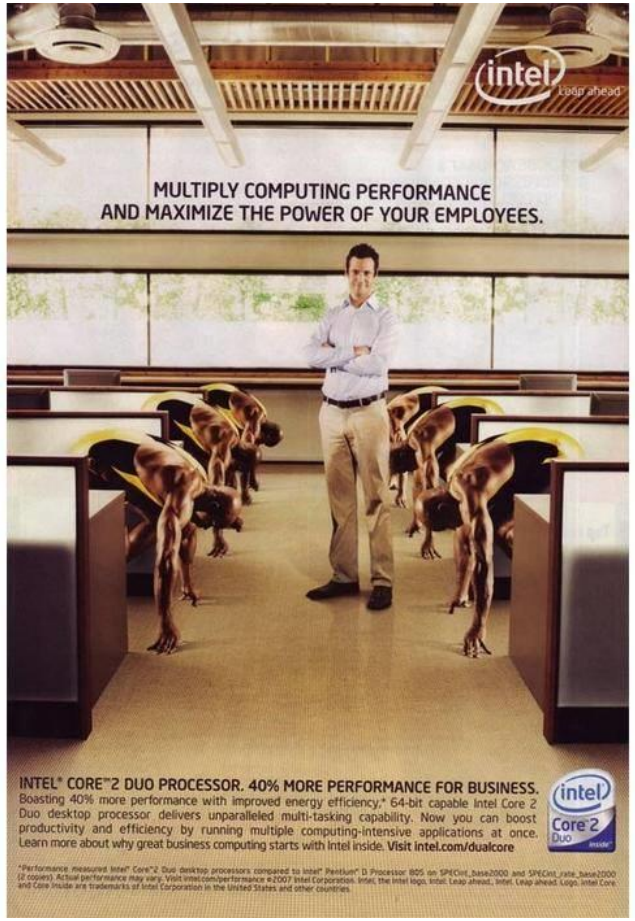
- **Material** foundation of AI hardware (servers, GPUs, sensors, batteries, etc.)
  - Cobalt, lithium, rare earth
- Rapid obsolescence and **e-waste** disposal and export
- AI as a material social-technical system deeply tied to histories of colonialism, capitalism, and inequality



Valdivia (2024)

The Case of Ads

Monsanto Chemicals (1950)



Intel (2007)

Dove (2011)

Presentation

- To ensure that class runs smoothly, please upload your ppt file to the BB by noon (12pm) on the day of your presentation. If your file is too large, submit a pdf version that includes an external download link.
- Q&A will take place in our class WeChat group. Please post your question there, and presenters will respond afterwards. Attendance on both March 4 & 11 is mandatory. Each student is required to ask at least one question during each session. This participation counts for 1×2 points (out of 20) in the presentation grade.